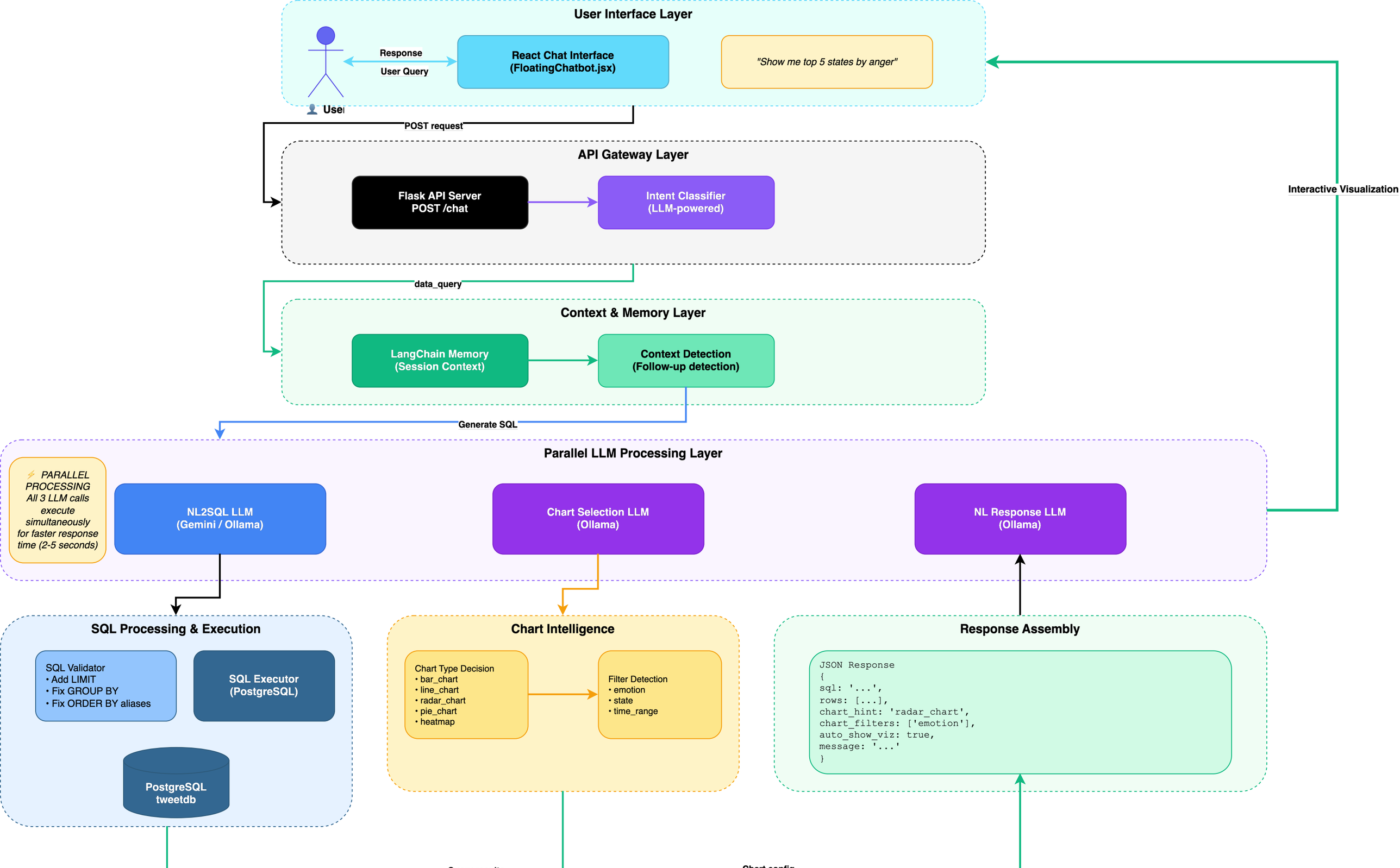
User Interface Layer
User
Response
User Query
React Chat Interface
(FloatingChatbot.jsx)
"Show me top 5 states by anger"
POST request
API Gateway Layer
Flask API Server
POST /chat
Intent Classifier
(LLM-powered)
Interactive Visualization
data_query
Context & Memory Layer
LangChain Memory
(Session Context)
Context Detection
(Follow-up detection)
Generate SQL
Parallel LLM Processing Layer
⚡ PARALLEL PROCESSING
All 3 LLM calls execute simultaneously for faster response time (2-5 seconds)
NL2SQL LLM
(Gemini / Ollama)
Chart Selection LLM
(Ollama)
NL Response LLM
(Ollama)
SQL Processing & Execution
SQL Validator
• Add LIMIT
• Fix GROUP BY
• Fix ORDER BY aliases
SQL Executor
(PostgreSQL)
PostgreSQL
tweetdb
Chart Intelligence
Chart Type Decision
• bar_chart
• line_chart
• radar_chart
• pie_chart
• heatmap
Filter Detection
• emotion
• state
• time_range
Response Assembly
JSON Response
{
sql: '...',
rows: [...],
chart_hint: 'radar_chart',
chart_filters: ['emotion'],
auto_show_viz: true,
message: '...'
}
Query results
Chart config